

DEVI AHILYA UNIVERSITY, INDORE

II BE (Civil Engineering)

Class Test III, NOV 2023

Subject Code: 3VLRC2

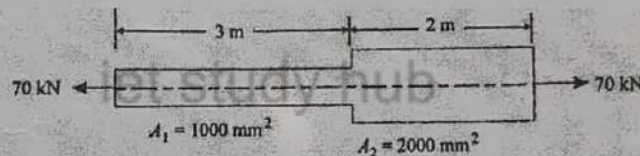
Subject Name: Applied Mechanics and Strength of materials

Duration: 70 Min

Maximum Marks: 20

Note: Attempt any four questions. All questions carry equal marks.

Q.1. A non-uniform tension member 5 m long is made up of two parts as shown in fig below. Find the total strain energy stored in the bar, when it is subjected to a gradual load of 70 kN. Also find the total strain energy stored in the bar, when the bar is made of uniform cross section of the same volume under the same load. Take $E = 2 \times 10^5 \text{ N/mm}^2$.



Q.2. Prove that stress produced by suddenly applied load is twice the one produced by the same load when applied gradually.

Q.3. State and prove Maxwell Reciprocal Theorem.

Q.4. State Castigliano's First Theorem. Using it, find the deflection of simply supported beam carrying a point load P eccentrically on the span. Assume uniform flexural rigidity.

Q.5. Derive an expression for Strain Energy Stored in a Body when the load is applying with impact.